

DSA Group Role-Play Report

Chapter 5: Tree Traversals — Pre-order, In-order and Post-order

Group Members

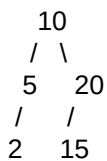
Name	ID Number
Fanuel Aregay	CEP0053/17
Abrham Nega	CEP0008/17
Aman Haftu	CEP0015/17
Habtamu Gizaw	CEP0064/17
Fikremaryam Belete	CEP0056/17

1. Chosen Topic and Rationale

Our group selected Tree Traversals from Chapter 5. We used a Binary Search Tree (BST) in our physical role-play simulation. The purpose was to demonstrate Pre-order, In-order and Post-order traversal and show the order in which nodes are visited.

2. BST Used in the Simulation

The root of our BST is **10**. The values used are **10, 20, 5, 2, 15**. Following the BST rule, smaller values are placed on the left and larger values on the right.



3. Traversal Rules and Results

Traversal	Rule	Visiting Order
Pre-order	Root → Left → Right	10 → 5 → 2 → 20 → 15
In-order	Left → Root → Right	2 → 5 → 10 → 15 → 20
Post-order	Left → Right → Root	2 → 5 → 15 → 20 → 10

4. Step-by-Step Operation Log

Traversal	Step	Visited Node
Pre-order	1	10
Pre-order	2	5
Pre-order	3	2
Pre-order	4	20
Pre-order	5	15
In-order	1	2
In-order	2	5
In-order	3	10

Traversal	Step	Visited Node
In-order	4	15
In-order	5	20
Post-order	1	2
Post-order	2	5
Post-order	3	15
Post-order	4	20
Post-order	5	10

5. Operation Tally

Pre-order: 5 node visits — 10, 5, 2, 20, 15.

In-order: 5 node visits — 2, 5, 10, 15, 20.

Post-order: 5 node visits — 2, 5, 15, 20, 10.

Total node visits across all three traversals: 15.

6. Critical Reflection: Limitations of Human Role-Play

- Remembering the exact traversal rule while physically moving can be difficult.
- Returning from a child node to its parent is harder to demonstrate physically than in recursive code.
- Participants may accidentally move to the wrong node or change the intended order.
- Limited physical space can make the BST structure less clear.
- Human role-play does not directly show the recursive function-call stack used by a computer program.

7. Simulation Video

Google Drive video:

https://drive.google.com/file/d/1Qp3p3_CSUuEIMc50cF2BCIpESsm1HC-e/view?usp=drivesdk

8. Conclusion

Our physical simulation demonstrates Pre-order, In-order and Post-order traversal using a BST with root 10. The same tree produces different visiting sequences because each traversal follows a different rule. The role-play helped us visualize the order in which each node is visited.